



## SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences  
Chennai- 602105



**Student Name:** Malavathu Saiteja

**Reg. No.:** 192424034

**Course Code:** DSA0308

**Slot:** D

**Course Name:** DATA HANDLING AND VISUALIZATION

**Course Faculty:** Dr. K. Kumaragurabaran

### Project Title: Personalized Healthcare Recommendation through Clinical NLP

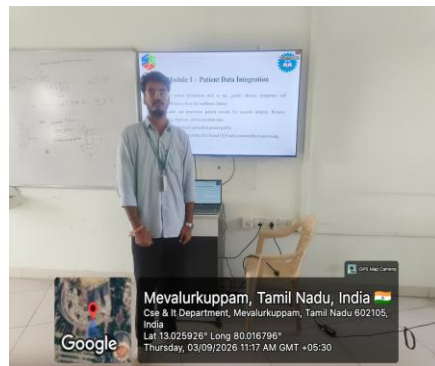
**Module Photographs:** (3 photographs – Module Photo, Individual student contribution module work in the project and presentation image)



This module prepares patient healthcare data for analysis. Patient information such as age, gender, disease, symptoms, and medical history is collected and checked for missing values, duplicate entries, and inconsistent records. Data cleaning and preprocessing produce a structured, unified patient profile ready for Clinical NLP processing.

**Main Outputs:**

- Cleaned, structured patient profiles
- Consistent, duplicate-free patient records
- Clean patient data ready for NLP and recommendation processing



### Project Description:

The **Personalized Healthcare Recommendation through Clinical NLP – Visualization & Analysis Module** is designed to present processed clinical and patient data in an interactive and meaningful way for improved healthcare insights and decision-making. The module utilizes cleaned and structured medical data, including patient history, symptoms, diagnoses, and treatment records, to generate visualizations such as trend charts, distribution plots, correlation heatmaps, and patient risk dashboards. Libraries like Matplotlib, Seaborn, Plotly, and other NLP visualization tools are used to create both static and interactive outputs for better interpretation of clinical data. Time-series analysis is applied to monitor patient health progression, treatment effectiveness, and disease patterns over time. NLP-based visualization techniques highlight key medical terms, symptom frequencies, and relationships between conditions extracted from clinical text. Comparative analysis is performed on different patient groups, diseases, and treatment outcomes to understand effectiveness and risks. The module provides insights such as disease prevalence, patient risk levels, treatment recommendations, and health trend patterns. Interactive dashboards enable users to filter information based on patient demographics, medical conditions, and time periods, making the system efficient, user-friendly, and valuable for personalized healthcare recommendations.

**Student Signature**

**Guide Signature**